

Equations with Parentheses

Name
Date
Course

The Distributive Rule

$$A(B+C) = AB+AC$$

Example 1: $2(3+6) = 2(3) + 2(6) = 6 + 12 = \boxed{18}$

Example 2: $-5(3x-4) = -5(3x+(-4)) = -5(3x) + (-5)(-4) = \boxed{-15x+20}$

① $2(x+4)-3=7+3(3x+2)$ ② $-4(2y+3)+5=6(y-2)-8$

$$2x+8-3=7+9x+6$$

$$-8y-12+5=6y-12-8$$

$$2x+5=13+9x$$

$$-8y-7=6y-20$$

$$-8=7x$$

$$\boxed{-\frac{8}{7}=x}$$

$$13=14y$$

$$\boxed{\frac{13}{14}=y}$$

③ $8+2(2z+1)=-3(z-4)-5$ ④ $-2(5x-3)=4(2x-1)$

$$8+4z+2=-3z+12-5$$

$$-10x+6=8x-4$$

$$4z+10=-3z+7$$

$$10=18x$$

$$7z=-3$$

$$\boxed{z=-\frac{3}{7}}$$

$$\frac{10}{18}=x$$

$$\boxed{\frac{5}{9}=x}$$

⑤ $-2(y+1)=-7(2y-3)$ ⑥ $7(z+3)-2(3z-5)+8=4-4(2+3z)$

$$-2y-2=-14y+21$$

$$7z+21-6z+10+8=4-8-12z$$

$$12y=23$$

$$\boxed{y=\frac{23}{12}}$$

$$z+39=-4-12z$$

$$13z=-43$$

$$\boxed{z=-\frac{43}{13}}$$

⑦ $2(9x-5)=3x+1$

$$18x-10=3x+1$$

$$15x=11$$

$$\boxed{x=\frac{11}{15}}$$

⑧ $2-5(3y-3)=2(y-1)$

$$2-15y+15=2y-2$$

$$-15y+17=2y-2$$

$$19=17y$$

$$\boxed{\frac{19}{17}=y}$$

⑨ $4(z+2)-1=-2(-3z-4)$ ⑩ $9-2x+5(2x-3)=3x-6(3-2x)$

$$4z+8-1=6z+8$$

$$9-2x+10x-15=3x-18+12x$$

$$4z+7=6z+8$$

$$8x-6=15x-18$$

$$-1=2z$$

$$\boxed{-\frac{1}{2}=z}$$

$$12=7x$$

$$\boxed{\frac{12}{7}=x}$$

$$\begin{array}{l}
 \textcircled{11} \quad 4x + 4(7x-1) + 2 = -2(x-5) \\
 4x + 28x - 4 + 2 = -2x + 10 \\
 32x - 2 = -2x + 10 \\
 34x = 12 \\
 x = \frac{12}{34} \\
 \boxed{x = \frac{6}{17}}
 \end{array}
 \quad
 \begin{array}{l}
 \textcircled{12} \quad 7 - 3(2z-6) = -2z-1 \\
 7 - 6z + 18 = -2z-1 \\
 -6z + 25 = -2z-1 \\
 -4z = -26 \\
 z = \frac{-26}{-4} \\
 \boxed{z = \frac{13}{2}}
 \end{array}$$

$$\begin{array}{l}
 \textcircled{13} \quad 5(-3x+2) = -2(2x+9) - 10(x-4) \\
 -15x + 10 = -4x - 18 - 10x + 40 \\
 -15x + 10 = -14x + 22 \\
 \boxed{-12 = x}
 \end{array}
 \quad
 \begin{array}{l}
 \textcircled{14} \quad -1 + 2(z-1) = 3(z+1) \\
 -1 + 2z - 2 = 3z + 3 \\
 2z - 3 = 3z + 3 \\
 \boxed{-6 = z}
 \end{array}$$

$$\begin{array}{l}
 \textcircled{15} \quad -2(x+2) + 4(2x-3) = 3x-8x \\
 -2x - 4 + 8x - 12 = 3x - 8x \\
 6x - 16 = -5x \\
 -16 = -11x \\
 \boxed{\frac{16}{11} = x}
 \end{array}
 \quad
 \begin{array}{l}
 \textcircled{16} \quad -(3y-1) + 7 = 8-y \\
 -1(3y-1) + 7 = 8-y \\
 -3y + 1 + 7 = 8-y \\
 -3y + 8 = 8-y \\
 0 = 2y \\
 \boxed{0 = y}
 \end{array}$$

$$\begin{array}{l}
 \textcircled{17} \quad 2 - (z+4) + 1 = -2(3z-5) \\
 2 - 1(z+4) + 1 = -2(3z-5) \\
 2 - z - 4 + 1 = -6z + 10 \\
 -z - 1 = -6z + 10 \\
 5z = 11 \\
 \boxed{z = \frac{11}{5}}
 \end{array}
 \quad
 \begin{array}{l}
 \textcircled{18} \quad -(2x+1) - 3 = 4 - (x-3) \\
 -1(2x+1) - 3 = 4 - 1(x-3) \\
 -2x - 1 - 3 = 4 - x + 3 \\
 -2x - 4 = 7 - x \\
 \boxed{-11 = x}
 \end{array}$$

$$\begin{array}{l}
 \textcircled{19} \quad 2(-3x+1) = -4(-x-3) \\
 -6x + 2 = 4x + 12 \\
 -10 = 10x \\
 \boxed{-1 = x}
 \end{array}
 \quad
 \begin{array}{l}
 \textcircled{20} \quad 3 - 4(-2x+4) = 9x-1 \\
 3 + 8x - 16 = 9x - 1 \\
 8x - 13 = 9x - 1 \\
 \boxed{-12 = x}
 \end{array}$$

$$\begin{array}{l}
 \textcircled{21} \quad -1 - (-7y+7) = 6(-5y-2) \\
 -1 - 1(-7y+7) = 6(-5y-2) \\
 -1 + 7y - 7 = -30y - 12 \\
 7y - 8 = -30y - 12 \\
 37y = -4 \\
 \boxed{y = -\frac{4}{37}}
 \end{array}
 \quad
 \begin{array}{l}
 \textcircled{22} \quad -2(3-4x) = 3(1+2x) \\
 -6 + 8x = 3 + 6x \\
 2x = 9 \\
 \boxed{x = \frac{9}{2}}
 \end{array}$$

$$\textcircled{23} -5 + 3(3-2z) = 4 - 2(4+z) \quad \textcircled{24} 2(-4-x) = -3(-1-2x)$$

$$-5 + 9 - 6z = 4 - 8 - 2z$$

$$-8 - 2x = 3 + 6x$$

$$4 - 6z = -4 - 2z$$

$$-11 = 8x$$

$$8 = 4z$$

$$\boxed{-\frac{11}{8} = x}$$

$$\boxed{2 = z}$$

$$\textcircled{25} -(2-3y) + 3y = 3 - 4(-2y+2) \quad \textcircled{26} -2(-5-4z) = -(z+1)$$

$$-2 + 3y + 3y = 3 + 8y - 8$$

$$10 + 8z = -z - 1$$

$$-2 + 6y = -5 + 8y$$

$$9z = -11$$

$$3 = 2y$$

$$\boxed{z = -\frac{11}{9}}$$

$$\boxed{\frac{3}{2} = y}$$

$$\textcircled{27} 2 - 4(3z+2) = -3(-1+2z) + 3(-2+4z) \quad \textcircled{28} 4x - (2-x) = 3$$

$$2 - 12z - 8 = 3 - 6z - 6 + 12z$$

$$4x - 2 + x = 3$$

$$-12z - 6 = -3 + 6z$$

$$5x - 2 = 3$$

$$-3 = 18z$$

$$5x = 5$$

$$-\frac{3}{18} = z$$

$$\boxed{x = 1}$$

$$\boxed{-\frac{1}{6} = z}$$

$$\textcircled{29} 4x - 2(3x-1) - 2x + 5x = 1 \quad \textcircled{30} 2 - 3x = 4 - (2-x) + x$$

$$4x - 6x + 2 - 2x + 5x = 1$$

$$2 - 3x = 4 - 2 + x + x$$

$$x + 2 = 1$$

$$2 - 3x = 2 + 2x$$

$$\boxed{x = -1}$$

$$0 = 5x$$

$$\boxed{0 = x}$$

$$\textcircled{31} 3 + 4x + 2x = -2(x-6) + 3(-6x+1) \quad \textcircled{32} -(x+5) - (2x-3) = 4-1$$

$$3 + 6x = -2x + 12 - 18x + 3$$

$$-x - 5 - 2x + 3 = 3$$

$$3 + 6x = -20x + 15$$

$$-3x - 2 = 3$$

$$26x = 12$$

$$-3x = 5$$

$$x = \frac{12}{26}$$

$$\boxed{x = -\frac{5}{3}}$$

$$\boxed{x = \frac{6}{13}}$$

$$\textcircled{33} -7 + 2(3+2x) = 4(-1+x) - 3(-4x+2) \quad \textcircled{34} 2 + 3x = 4 - (8-x)$$

$$-7 + 6 + 4x = -4 + 4x + 12x - 6$$

$$2 + 3x = 4 - 8 + x$$

$$-1 + 4x = 16x - 10$$

$$2 + 3x = -4 + x$$

$$9 = 12x$$

$$2x = -6$$

$$\frac{9}{12} = x$$

$$\boxed{x = -3}$$

$$\boxed{\frac{3}{4} = x}$$